D2 Detailed Design — Smart Closet Mirror

Assignment #4 (Design Diagrams), part D2 · Team bullriders · Arya Garg, Clare Hooper, Rehman Khalid, Jace Schubert · Week 7

1. Header, conventions, and D2 overview

Project title: Smart Closet Mirror

> **Goal statement.** Turn a Gen 1 Mirror into a closet mirror that measures the user with a Kinect, shows their own clothes and shop items on their body, and tells them what to wear today and whether a new piece is worth buying.

Overview for a reviewer with no context (two minutes)

What the system is. A Gen 1 Mirror (an Android 6 smart mirror) becomes the screen for a closet assistant. A Kinect 360 on top of the Mirror sees the user in colour and depth. A Ubuntu PC in the home (the "brain") does all the work in eight layers, L1–L8 (D0, Fig 1): capture, perception, body, wardrobe, try-on, decision, connectors to outside services, and presentation. The Kinect stays off until the person at the glass allows it for that session. The user adds clothes by photographing them with a phone. The Mirror shows them on the user's body in live video and tells them what to wear today and whether a shop item is worth buying.

What D1 covered. D1 detailed the two riskiest components. L4 Wardrobe has the main data model (D1 Fig 1) and the phone API. L5 Try-on has the fit check (algorithm A1) and the live garment drawing (algorithm A2).

What D2 adds.

- **Changes driven by the Week 4 user stories**, marked [US-xx] where they appear: current weather in outfit suggestions (US-01), camera capture only after consent (US-04), a profile PIN so each person sees only their own wardrobe (US-05), and configuration rollback for maintainers (US-03).
- Every remaining component (L1, L2, L3, L6, L7, L8) and the shared planes (data store, message bus) now has a data model, behaviour, and interface (part 2).
- The four glue components get the same treatment: authentication, notifications, configuration, and logging and health.
- Three state machines: the Mirror session (Fig D2-1), body measurement (Fig D2-2), and camera consent (Fig D2-5).
- Two flows that show their error paths: "buy or skip?" (Fig D2-3) and "recommend an outfit" (Fig D2-6).
- The data model for the new tables (Fig D2-4).
- Revisions to five D0 interfaces that had grown too loose (part 2, first table).
- The edge-case table, requirements traceability, design rationale, and peer-review notes (parts 3–6).

Known gaps and open questions

#	Gap or question	Current position

G1	No shop product API or image search provider has been chosen yet (I14b, I14c).	L7 uses one adapter per provider, so the choice does not affect other layers. Candidates must allow non-commercial student use without fees. **Decided:** images come from the Wikimedia Commons API (free, no key, openly licensed images). Products come from a local catalogue adapter that reads hand-entered listings from a JSON file, because no free shop API we found allows this use. Owner: Jace Schubert. A7, the only heavy user of both, is now stretch scope (G9).
G2	A profile PIN typed on the glass can be watched by someone standing nearby (US-05).	The keypad shuffles its digits on every unlock and does not echo them. The PIN protects wardrobe privacy, not payments. Unlocking from the paired phone is a possible later improvement.
G3	Accuracy of body circumferences from one front and one side depth view (A3) is unknown.	Target ±2 cm against a tape measure on all four team members before Week 9. The calibration factors live in the configuration file.
G4	Phone notifications are in-app only (no push), because push services would send data off the home network.	The phone page polls when open; the glass shows toasts. Accepted as a privacy trade-off.
G5	Which machine is the brain PC, and does it have an NVIDIA GPU (needed for A2 and NVENC)?	**Confirmed:** the brain PC is Clare Hooper's Ubuntu desktop with an NVIDIA GPU (D1 §6), so A2 and NVENC run on it. The libx264 software fallback at 20 fps stays in the design in case the NVIDIA driver fails (EC-L5-X).
G6	Kinect 360 depth quality under the Mirror's own lighting and at 1–3 m in a typical room.	To be tested in the Week 8 hardware spike.
G7	The traceability table (part 4) lists the five user stories but the mapping is not yet filled in.	**Closed:** part 4 maps US-01 to US-05 and NF-01 to NF-04 to design elements and edge cases, and the reverse check covers every component.
G8	The Week 4 user stories call the project "Smart Mirror Personal Stylist"; D0–D2 use "Smart Closet Mirror".	**Decided:** "Smart Closet Mirror" in every document and diagram (review item PR-03). User_Stories takes the new title in its next revision.
G9	The user stories do not mention try-on, body measurement, buy-or-skip, or looks from photos.	**Decided:** L3, L5, A3, and A5 stay. They carry the goal statement (measure the user, show clothes on their body, say whether a new piece is worth buying) and are traced to NF-04 and the goal in part 4. A7 (look from a photo) moves to stretch scope (review item PR-02, rationale R9): it stays designed but is built only if the core is finished by Week 11.

Conventions for every D2 figure

Each figure carries the project title, the goal statement, and its own legend. The legend on the figure is the authority; this table repeats it. D0 and D1 figures keep the conventions stated in those documents.

Figure	Boxes and shapes	Lines
Fig D2-1 Mirror session (state machine) Fig D2-2 Body measurement (state machine) Fig D2-5 Camera consent (state machine)	Rounded box = state. Dashed square box = error state. Thick box = final (success) state. Black dot = start state. Large dashed rounded outline = a group of states.	Arrow = transition, labelled "trigger" or "trigger / action"; text in [brackets] is a guard that must be true. Dashed arrow = error or timeout transition. An arrow leaving a group applies to every state inside it.
Fig D2-3 "Buy or skip?" with error paths (flow) Fig D2-6 "Recommend an outfit" with error paths (flow)	Rounded box = step, naming the component and interface. Diamond = check. Thick box = successful end. Dashed box = error outcome, naming the decided behaviour (retry, degrade, fall back, surface error) and the handler.	Solid arrow = happy path. Dashed arrow = error path. Labels on arrows are check outcomes (yes / no).
Fig D2-4 Data model (entity-relationship)	Box = entity (table); rows = key attributes with types. Dashed box = entity defined in D1 Fig 1. Keys: PK = primary key (underlined), FK = foreign key, UQ = unique, UQ1 = unique together, IX = indexed.	Line = relationship; the italic verb reads from the first-named entity. Cardinality at both ends in crow's-foot notation, repeated as text: 1, 0..1, 1..*, 0..*.

Interface IDs continue D0's numbering (I1–I17). D2 adds I18–I28 and splits I14 into I14a–d (see "Interface revisions"). Algorithm IDs continue D1's: A1–A2 in D1, A3–A7 here. Edge-case IDs are EC-<component>-<category>, for example EC-L7-X.

2. Finished components (the D2 diagrams)

D1 already finished L4 Wardrobe and L5 Try-on. This part finishes everything else. Each component lists its **data model**, its **behaviour**, and its **interface** with errors. Every error, on the bus or over HTTP, uses the D1 shape {"error": {"code", "message", "fields"}}.

Interface revisions (D0 → D2)

ID	D0 definition	Problem	D2 revision
I6 → I6 + I18	L8 → WebView: ui_state and control on one channel.	Mixed two concerns: screen content (idempotent, resendable) and device commands (not idempotent, such as waking the glass).	**I6 ui.state** (WebSocket, full state with seq) and **I18 mirror.control** (JS bridge, one command per message with an ack).
I7	FrameBundle with raw shared-memory handles.	Leaked L1's memory layout; a consumer could read a buffer L1 was overwriting.	frame_bundle/2: ring slot 0–3 plus a generation counter. Consumers check the generation before and after reading and discard the frame if it changed.
I9	Avatar {mesh_ref glTF, rig}.	After D1 moved rendering to a 2-D warp (A2), nothing reads a 3-D mesh; the mesh was an internal L3 detail.	**BodyModel** {profile_id, version, rest_joints, proportions}, the only fields A1 and A2 use.
I14 → I14a–d	One ContextQuery{kind, params: map}.	Too loose: an untyped map meant callers and L7 could disagree on parameters, and errors could not be specific. It also had no weather, which US-01 needs.	Four typed methods: **I14a ctx.calendar_day**, **I14b ctx.products**, **I14c ctx.images**, **I14d ctx.weather** [US-01].
I5	Mirror camera + mic → L8: presence at 2 Hz and voice commands.	Presence came from watching the room through the Mirror's camera before anyone agreed, which US-04 forbids.	I5 carries only touch events and wake-word or voice commands. The Mirror's camera is never used [US-04].
I15 → I15 + I19	L6 ↔ L2 carried image analysis *and* gesture events.	Two different timings (a 5 s request vs a stream of events) and directions on one ID.	**I15 vision.analyze** (request/reply) and **I19 intent.gesture** (L2 → L6 events).
I8	BodyMeasurements.	L3 needed the T-pose joint positions to build the BodyModel and had no way to get them.	Adds rest_joints (20 × xyz, metres). Schema body_measurements/2.
New	Glue and control interfaces that D0 did not draw.		**I20** notify.create · **I21** ops.health · **I22** pairing API · **I23** config.changed · **I24** measure.start / cancel (L6 → L2) · **I25** measure.progress (L2 → L8) · **I26** notifications API (phone) · **I27** capture.enable / disable (L8 → L1) [US-04] · **I28** auth.unlock (L8 → Auth) [US-05].
D1 entities	Additions to D1 Fig 1 (D1 itself is unchanged; this table records them).		**Garment** gains warmth: INT 1–3 and water_resistant: BOOL, set when a photo is uploaded and editable by PATCH; A4 needs them [US-01]. **GET /api/v1/garments**: the optional owner parameter now accepts only the caller's own id; any other returns 403 FORBIDDEN [US-05].

L1 Capture — owner: Rehman Khalid

Data model: no persistent data. It holds only a 4-slot ring buffer in shared memory and its device settings from configuration.

Behaviour (steps; the device states are Disabled → Opening → Streaming → Stalled, with Disabled the default):

1. **Consent gate [US-04]:** start in Disabled with the Kinect closed and its LED off. Open the device only when L8 sends capture.enable (I27) for a session where the user said yes (Fig D2-5). On capture.disable, or when the session ends, close the device, turn the LED off, and confirm. Frames stay in shared memory; L1 has no code path that writes them to disk.
2. Open the Kinect with libfreenect (device index from config). Set RGB to 640 × 480 and depth to "registered" mode, so depth pixels line up with colour pixels.
3. The driver calls back once per RGB frame and once per depth frame. Pair the two when their timestamps are within 20 ms, and write the pair into the next ring slot. Increment that slot's generation.
4. Publish a FrameBundle (I7) with the slot number and generation. Every 5 s, publish health (I21) with the measured frame rate.
5. If no pair arrives for 2 s, go to Stalled: publish health DOWN, close the device, and try to reopen it every 5 s. When frames flow again, publish health OK.

Interface	Input	Output	Errors
I7 frames.raw (publishes)	—	frame_bundle/2 {seq: uint64, ts: int64 ns, slot: int 0–3, generation: uint64, width: 640, height: 480}	None sent; a consumer that sees the generation change discards the frame.
capture.status (bus request)	{}	{state: enum{disabled, opening, streaming, stalled}, device_open: bool, fps: float, last_frame_ts: int64 ns}	NO_SERVICE (bus) if L1 is not running; L8 treats this as "Camera status unknown".
I27 capture.enable / capture.disable (bus request, from L8)	{session_id: UUID, required; consent_id: UUID, required for enable}	{state: "streaming" | "disabled", device_open: bool}, sent only after L1 has checked the device	NO_CONSENT — consent_id unknown or already revoked · DEVICE_NOT_FOUND · TIMEOUT (3 s at the caller → Unknown, Fig D2-5)
I21 ops.health (publishes)	—	See Logging and health.	DEVICE_NOT_FOUND and USB_TRANSFER_ERROR are reported as detail, with state DOWN.

L2 Perception — owner: Arya Garg

Data model: no persistent data. Model weights are files listed in configuration. Measurements go to L3 (I8), which stores them.

Behaviour. L2 only receives frames while capture is on (Fig D2-5), so pose, gestures, and measurement are unavailable when the user has declined [US-04]. L2 runs four jobs in one process:

- **Pose (every frame, ≤ 60 ms, D0 budget):** run MediaPipe Pose on the RGB frame to get 33 landmarks, and map them to our 20 joints. Each joint's z is the median depth in a 5 × 5 window around it, ignoring zero pixels. Mask = MediaPipe segmentation AND depth within ±0.6 m of the torso. Publish I10, newest frame only.
- **Gestures:** rules over the smoothed joints.
  - A swipe is the right wrist moving more than 0.35 m sideways within 0.5 s at chest height: *next* or *back*.
  - A right hand held above the head for 1 s is *select*.
  - Only gestures with confidence ≥ 0.8 are published, at most one per second (I19).
- **Measurement:** algorithm A3, below and in Fig D2-2.
- **Item detection (I15):** run an open-vocabulary detector (OWLv2) with one text prompt per garment category. Keep boxes with confidence ≥ 0.3, and merge overlaps (IoU > 0.5). The dominant colour is k-means (k = 3) on the box pixels, named by the nearest of 16 palette colours in CIELAB.

Algorithm A3: body measurement from T-pose (state machine in Fig D2-2).

- **What it is:** it turns about 30 front and 30 side depth frames into body measurements in centimetres.
- **In:** FrameBundles plus LivePose for one user, with user_id and session_id.
- **Out:** BodyMeasurements (I8).
- **Steps:**
  1. Fit the floor plane once with RANSAC (200 iterations) on depth points below the hips.
  2. Height = distance from the floor to the highest mask point.
  3. At the chest, waist, and hip levels (set from the shoulder and hip joints), take the front width *w* = mask row width × depth / fx, and the side depth *d* the same way from the side frames.
  4. Circumference = the perimeter of an ellipse with axes *w* and *d* (Ramanujan's formula), times a calibration factor from configuration (gap G3).
  5. Shoulder width comes from the front mask at shoulder height. Inseam = hip-joint height above the floor minus a calibrated offset.
  6. Each value is the median across frames.
  7. Confidence = mean joint confidence × (1 − mean relative spread across frames), limited to 0–1.
- **Complexity:** O(P) per frame for P = 307,200 pixels, over about 60 frames, which takes under 1 s in total.

Interface	Input	Output	Errors
I24 measure.start / measure.cancel (request)	{schema: "measure_start/1", user_id: UUID, required; session_id: UUID, required}	{accepted: true, state: "wait_front"}	BUSY — a session is already running · NO_PERSON — nobody in view at start

I25 measure.progress (publishes)	—	{session_id, state: enum (Fig D2-2), hint: string ≤ 60, progress: float 0–1, reason: string?}	On failure, state = failed and reason ∈ {TIMEOUT, PERSON_LOST, MULTIPLE_PEOPLE, LOW_CONFIDENCE, OUT_OF_RANGE}.
I8 body.measured (publishes)	—	body_measurements/2: D0 fields plus rest_joints: 20 × {x, y, z: float m}	Only published when confidence ≥ 0.7.
I10 body.pose, **I19** intent.gesture (publishes)	—	I10 as in D0. I19 {user_id, intent: enum{next, back, select}, conf: float, ts}	None; when tracking is lost, I10 simply has no person.
I15 vision.analyze (request)	{request_id: UUID; image_refs: URI[], 1–6 items}	DetectedItems as in D0, plus color_name: string	TOO_MANY_IMAGES (> 6) · IMAGE_UNREADABLE (per image, in fields) · MODEL_UNAVAILABLE — weights missing or GPU out of memory · timeout 5 s at the caller

L3 Body — owner: Arya Garg

Data model: BodyProfile (D1 Fig 1) and BodyModel (Fig D2-4). Each profile has 0..* model versions, one per algorithm version.

Behaviour (steps on each I8 message):

1. Check the message again (defence in depth): confidence ≥ 0.7 and every value within human range. If either fails, log and drop it.
2. If the user's current profile differs by less than 1.0 cm in every measurement, keep it. Only update captured_at, so the fit cache stays valid.
3. Otherwise, in one transaction: insert a new BodyProfile, clear is_current on the old one, and set it on the new one.
4. Build the BodyModel: rest joints from I8, and proportions (torso, arm, and leg length; shoulder width in cm) from the joints.
5. Publish I9. L5's fit cache misses on its own, because the profile_id is new.

Interface	Input	Output	Errors
I9 body.model (publishes)	—	body_model/1 {user_id, profile_id, version: int, rest_joints: 20 × xyz m, proportions: {torso_cm, arm_cm, leg_cm, shoulder_cm}}	—
body.current (request)	{user_id: UUID, required}	The latest body_model/1 for the current profile	NO_CURRENT_PROFILE — the user has never been measured · UNKNOWN_USER

L6 Decision — owner: Clare Hooper

Data model: Outfit, OutfitItem, and WearEvent (D1 Fig 1), plus Decision, StylePreference, and SpendEntry (Fig D2-4). StylePreference holds the home location, rounded to 0.1°, for weather. Every query is scoped to the unlocked user [US-05].

- Decision keeps garment_ids as a JSON snapshot, so a past decision still reads correctly after a garment is archived.
- inputs_hash lets L6 reuse a decision when nothing has changed.
- Budget left this month = User.monthly_budget − the sum of this month's SpendEntry rows.

Behaviour. The five algorithm parts for A1/A2 are in D1. A4, A5, and A7 are given here as inputs, outputs, steps, and cost; their edge cases go in part 3.

- **A4 Outfit recommendation [US-01, UC-01]** (flow and error paths in Fig D2-6).
  - **In:**
    - occasion: from the request, else the first calendar event (I14a), else casual
    - closet index (I13) for the unlocked user only
    - weather (I14d)
    - StylePreference, WearEvents for the last 14 days, FitResults for the current profile
  - **Out:** a DecisionView of kind outfit_of_day with a top, a bottom (or a dress), shoes, and optional outerwear, plus a reason naming the occasion and the weather. Or "No complete outfit available" with the missing categories.
  - **Steps:**
    1. **Completeness first:** the available items must include a top and a bottom (or a dress), and shoes. If not, return "No complete outfit available" with the missing categories (for example, "bottom"). Never suggest an item that is not in the inventory (C-01.3).
    2. **Formality:** the target comes from the occasion (business 4, formal 5, casual 2, sport 1), plus formality_bias.
    3. **Weather:** use it only if observed_at is no more than 60 minutes old.
       - Feels-like below 5 °C → warmth 3, outerwear required. 5–15 °C → warmth ≥ 2, outerwear optional. Above 25 °C → warmth 1, no outerwear.
       - Chance of rain ≥ 50% → prefer water-resistant outerwear and shoes.
       - If the weather is missing or older than 60 min, skip these rules. The reason then says "Weather unavailable" and does not call the outfit weather-appropriate (C-01.2).
    4. **Shortlist:** for each category, keep garments that are available, not worn within repeat_window_days, and not tight. Score them: −|formality − target|, −|warmth − warmth target|, +1 for a favourite colour, −2 for an avoided colour. Keep the top 10.
    5. **Combine:** top × bottom (or a dress) × shoes, plus outerwear when required or optional. Combination score = sum of item scores + colour harmony (at most 2 non-neutral colours) + freshness (days since worn, capped at 14). Pick the highest score; on a tie, take the outfit worn least recently.
    6. **Reason (≤ 80 characters):** for example, "Class presentation, 8 °C and rain: wool coat, waterproof boots".
  - **Cost:** at most 10 × 10 × 10 × 10 = 10^4 combinations, under 50 ms. The weather call is cached for 15 min. L6 waits at most 3.5 s for I14d; after that, A4 continues as "Weather unavailable" while L7's retries refresh the cache for the next request, so the card still arrives within 5 s (review item PR-04).
- **A5 Buy-or-skip verdict.**
  - **In:** a shop garment and its listing, FitEvaluation (A1, or "pending"), closet index, budget left.

- **Out:** verdict buy or skip, with up to 3 reasons. The rules are applied in order, and each one that fires adds a reason:
  1. No size is "good" → skip.
  2. Price is above the budget left → skip.
  3. The closet already has a near-duplicate (same category and colour name, formality ±1) → skip.
  4. The closet has fewer than 3 items in this category, or none at the event formality → buy ("fills a gap").
  5. Otherwise → buy if the fit is good and the price is within budget.
- **Cost:** O(closet) ≈ 300 comparisons.

- **A7 Look from a photo of a public figure.**
  1. Get images with ctx.images (I14c), up to 6.
  2. Detect pieces with vision.analyze (I15).
  3. For each category, keep the most frequent colour across the images.
  4. For each piece, find a closet match (same category, colour difference ΔE < 20). If there is none, search shop items with ctx.products (I14b), limited by category, colour, and max_price = budget left.

  Out: a DecisionView of kind look. **Cost:** dominated by I15, about 0.6 s per image.
- **Request handling:** a UserAction (I17) chooses the algorithm. "measure" calls I24. The result is saved as a Decision and sent to L8 as a DecisionView. Fig D2-3 shows A5's full path, including its errors.

Interface	Input	Output	Errors
I17 ui.action → decision.view	UserAction as in D0; action adds measure, ask_verdict, and recommend with occasion: enum, optional and location_name: string, optional (default: StylePreference location) [US-01]	DecisionView as in D0; reasons limited to 3	UNKNOWN_ACTION · STALE_VIEW (old view_id: L6 resends the current view)
GET /api/v1/decisions (phone)	kind: enum, optional · limit: int 1–50, default 20 · cursor: string, optional	200 {items: [{decision_id, kind, verdict, reasons, garment_ids, created_at}], next_cursor}	400 BAD_QUERY · 401 UNAUTHORIZED
POST /api/v1/spend (phone)	amount: decimal string > 0, required · currency: ISO 4217, required · listing_id: UUID, optional · spent_on: date, optional, default today	201 {spend_id, budget_left: decimal string}	400 VALIDATION_FAILED · 401 · 404 LISTING_NOT_FOUND
PUT /api/v1/preferences (phone)	StylePreference fields, all optional, with the ranges in Fig D2-4	200 StylePreference	400 VALIDATION_FAILED · 401

L7 Connectors — owner: Jace Schubert

Data model: ProviderAccount, CategoryRule, and CacheEntry (Fig D2-4). OAuth tokens are encrypted with a key kept in the secrets file, never in the database.

Behaviour (the same pipeline for every outside call):

1. Build the cache key (sha256 of provider + endpoint + normalised parameters). Return a fresh hit right away.
2. **Privacy filter:** each endpoint has an allowlist of outgoing parameters. Body data, images, and user ids never leave the house.
3. Take a token from the provider's rate-limit bucket (rate from configuration).
4. Call over HTTPS with a 3 s timeout. On 429, 5xx, or a timeout, retry after 1, 2, then 4 s, honouring Retry-After. On 401, refresh the token once.
5. Normalise the response into our models, cache it with a TTL (calendar 15 min, weather 15 min, products 6 h, images 7 days), and reply. Products the user opens are also sent to L4 (I16).

A6 Calendar categorisation.

- **In:** event title (string).
- **Out:** category ∈ {business, casual, formal, sport}.
- **Steps:** lower-case the title. Try the enabled CategoryRule patterns in priority order, user rules before default rules. The first match wins; no match → casual.
- **Default rules:** interview|client|presentation|meeting → business; wedding|gala|ceremony → formal; gym|run|yoga|match → sport.
- **Cost:** O(R) with R ≈ 30 rules.

Interface	Input	Output	Errors
I14a ctx.calendar_day	user_id: UUID, required · date: YYYY-MM-DD, required, today ± 7 days	{events: [{start, end: RFC 3339, title, category}], stale: bool, fetched_at}	NOT_LINKED · NEEDS_RELINK · PROVIDER_UNAVAILABLE (no cache) · BAD_QUERY
I14d ctx.weather [US-01]	lat, lon: float, required, rounded to 0.1° (about 10 km) before leaving the house	{temp_c, feels_like_c: float, precip_prob: int 0–100 %, condition: enum{clear, cloudy, rain, snow, storm, fog}, wind_kph: float, observed_at: RFC 3339, stale: bool}. Provider: Open-Meteo (no API key; free for non-commercial use).	PROVIDER_UNAVAILABLE — provider down and no cached reading · BAD_QUERY — coordinates out of range. A cached reading older than 60 min is still returned with stale = true; A4 treats it as "Weather unavailable".
I14b ctx.products	query: string 1–100, required · category, color, size_label: optional · max_price: decimal, optional · currency: ISO 4217 · limit: int 1–20, default 10	{products: [{sku, shop, title, price, currency, url, sizes, image_ref}], stale, fetched_at}	PROVIDER_UNAVAILABLE · RATE_LIMITED (local bucket empty for > 3 s) · BAD_QUERY
I14c ctx.images	query: person name, 1–60 chars, required · limit: int 1–6, default 6	{image_refs: URI[] (downloaded into the data store), stale, fetched_at}	PROVIDER_UNAVAILABLE · RATE_LIMITED · BAD_QUERY
Provider linking (phone)	POST /api/v1/providers/{provider}/link · GET …/callback?code&state · DELETE …/{provider}	201 {auth_url} · 302 to the phone page after the callback · 204	400 UNKNOWN_PROVIDER · 400 BAD_STATE (CSRF state mismatch) · 502 PROVIDER_REFUSED

L8 Presentation — owner: all (lead: Clare Hooper)

Data model: ConsentEvent (Fig D2-4). It records each allow, decline, or revoke with a session id but no user link and no image [US-04]. Session state lives in memory and is rebuilt from L6 after a restart; screen text and layout come from configuration.

Behaviour: the state machine in Fig D2-1, plus these steps:

1. When the Mirror connects, check its device token (Auth), then send the full ui.state (I6).
2. Each state change sends a new full state with seq + 1.
3. Voice commands (I5) are matched against a fixed list: next, back, select, buy, skip, try, measure, recommend, camera off, sleep. A command is accepted when its edit distance is ≤ 1 and confidence ≥ 0.6, and it becomes a UserAction (I17).
4. Health events (I21) and measurement progress (I25) update the banner and the T-pose guide.
5. **Camera consent [US-04]** (Fig D2-5):
   - The glass always shows "Camera off" or "Camera on".
   - Before try-on, measuring, or gestures, if capture is off, L8 asks "Allow the camera for this session?".
   - **Allow:** L8 sends capture.enable (I27) and shows "Camera on" only after L1 confirms.
   - **Decline, or no answer in 15 s:** L8 logs a ConsentEvent and leaves the camera off.
   - **Revoke:** "Camera off" (touch or voice) sends capture.disable.
   - While awake, L8 checks capture.status every 2 s. No reply, or a state that contradicts what L8 expects, means "Camera status unknown", and image functions are blocked (C-02.2).
6. **Profile unlock [US-05]:** after the user picks a profile, the keypad (digits shuffled) sends auth.unlock (I28). The session is bound to that user_id, and only that user's wardrobe, cards, and toasts are shown. Asleep ends the session.

Interface	Input	Output	Errors
I5 (in, WebSocket)	touch {x, y: int px, ts} · wake_word {ts} · voice_command {text ≤ 40, confidence 0–1}. No camera data [US-04].	—	Close 4401 — bad device token · 4409 — a second Mirror connection replaced this one
I6 ui.state (out, WebSocket)	—	ui_state/2 {seq: int, screen: enum (Fig D2-1 states), cards: Card[] ≤ 4, banner: {kind, text} | null, toast: string | null}	Close 4500 — server error; the Mirror reconnects every 2 s
I27 capture.enable / disable (out, to L1)	—	See L1	TIMEOUT or NO_SERVICE → state Unknown, "Camera status unknown" (Fig D2-5)
I18 mirror.control (out, JS bridge)	—	{cmd: enum{wake, sleep, brightness}, value: int 0–100?, cmd_id: UUID}; the Mirror replies {cmd_id, ok: bool}	No reply in 1 s: retry once, then log (glass state unknown)

Planes: data store and message bus — owner: Rehman Khalid

Data store — data model: the union of D1 Fig 1 and Fig D2-4, in one SQLite file (brain.db), with images under files/<garment_id>/.

Data store — behaviour:

- Settings: WAL mode, foreign_keys = ON, busy_timeout 2,000 ms.
- **Each table has exactly one writing component** (the owners in Fig D2-4 and D1); others read only.
- Schema migrations (Alembic) run at start-up.
- Nightly at 03:00: a .backup of the database plus a copy of the files to an external drive, keeping 7 copies.

Data store — interface: one repository module per owning component (Python functions; no shared SQL strings).

- **StorageBusy** (still locked after 2 s): callers map it to HTTP 503 or the bus error STORAGE_UNAVAILABLE.
- **StorageFull** (under 1 GB free): the store goes read-only and health reports DEGRADED.

Message bus — data model: no persistent data.

Message bus — behaviour: one broker process, busd. A ZeroMQ XSUB/XPUB proxy handles events, and a ROUTER socket handles request/reply by service name. Heartbeats every 1 s.

Message bus — interface:

- Every message uses one envelope: {schema: "name/version", msg_id: UUID, corr_id: UUID | null, sender: string, ts: int64 ns, body: object}.
- publish(topic, body) and request(service, body, timeout_ms) → body.
- Errors: TIMEOUT · NO_SERVICE (no worker registered) · TOO_LARGE (> 1 MB; send a reference instead) · BAD_SCHEMA (unknown schema name or version).

Glue: Authentication — owner: Jace Schubert

Data model: Device and ProfileLock (Fig D2-4).

Behaviour (pairing steps):

1. On the glass, the user picks "pair phone". L8 asks Auth for a 6-digit code, valid for 120 s with at most 5 attempts.
2. The phone opens /pair and enters the code.
3. Auth creates a Device and returns a token dv_<device_id>.<32 random bytes, base64url>. The token is shown only once; only its argon2id hash is stored.
4. On each request, the device_id prefix finds the row and the hash is verified. Revoked devices fail.

The Mirror's own token is provisioned once at install.

Profile unlock and wardrobe separation [US-05] (steps):

1. Each user sets a 4–6 digit PIN from their own paired phone. Only its argon2id hash is stored, in ProfileLock.
2. On the glass, auth.unlock (I28) checks the PIN. It returns a session bound to that user_id that lasts until Asleep. After 5 wrong PINs the profile is locked for 5 min.
3. **Isolation rule:** every repository query takes user_id from the authenticated session or device token, never from the request body. Asking for another user's data returns 403 FORBIDDEN (HTTP) or the bus error FORBIDDEN.

Interface	Input	Output	Errors
auth.pair_code (bus, from L8)	{user_id: UUID, required}	{code: 6 digits, expires_at}	UNKNOWN_USER
I28 auth.unlock (bus, from L8)	{user_id: UUID, required; pin: 4–6 digits, required}	{session_id: UUID, user_id, expires_at}	WRONG_PIN (remaining attempts in fields) · LOCKED (until locked_until) · NO_PIN_SET — the user must set one from their phone
PUT /api/v1/profile/pin (phone)	pin: 4–6 digits, required; current_pin: required if a PIN is already set	204	400 VALIDATION_FAILED · 401 · 403 WRONG_CURRENT_PIN
I22 POST /api/v1/pair	code: 6 digits, required · device_name: 1–40 chars, required	201 {device_id: UUID, token: string (shown once), user_id}	404 CODE_UNKNOWN · 410 CODE_EXPIRED · 429 TOO_MANY_ATTEMPTS
GET / DELETE /api/v1/devices[/{id}]	Bearer token; id: UUID for delete	200 {items: [{device_id, kind, name, last_seen_at}]} · 204	401 UNAUTHORIZED · 404 NOT_FOUND · 409 CANNOT_REVOKE_SELF
verify(token) (in-process)	token: string	Device	Unauthorized (→ HTTP 401 / WebSocket close 4401)

Glue: Notifications — owner: Clare Hooper

Data model: Notification (Fig D2-4).

Behaviour:

1. On notify.create, if an unread notification with the same (user_id, dedupe_key) exists, update it instead of adding a new one.
2. Publish notify.new; L8 shows a toast if that user is at the glass.
3. The phone page polls I26 every 60 s while it is open.
4. Expired rows are deleted daily.

The kinds are outfit_ready, relink_needed, measure_due (30 days after the last measurement), and system.

Interface	Input	Output	Errors
I20 notify.create (bus)	user_id: UUID, required · kind: enum, required · title: 1–60 chars · body: ≤ 200 chars · dedupe_key: string, optional · ttl_h: int 1–720, default 72	{notification_id: UUID, deduped: bool}	BAD_INPUT · UNKNOWN_USER
I26 GET /api/v1/notifications	unread: bool, optional · limit: 1–50, default 20 · cursor, optional	200 {items: [{notification_id, kind, title, body, created_at, read_at}], next_cursor}	400 BAD_QUERY · 401
POST /api/v1/notifications/{id}/read	id: UUID	204	401 · 404 NOT_FOUND

Glue: Configuration — owner: Rehman Khalid

Data model: no persistent data in the database; configuration must still work when the database is the broken part. Three things on disk:

- /etc/scm/config.yaml: in git as config.example.yaml.
- /etc/scm/secrets.env: mode 600 and never committed. It holds the OAuth client secrets and the token encryption key.
- /var/lib/scm/config_history/ [US-03]: a numbered copy of every configuration that loaded successfully, plus index.json with {version, sha256, applied_at, known_good: bool}.

Key groups: capture, perception thresholds, A1 ease bands, A3 calibration, A4 weights, L7 rates and TTLs, timeouts, UI text.

Behaviour:

- **Start-up:** each process loads both files and validates them against a typed schema (Pydantic). If anything is invalid, it exits with a message naming the bad key, so systemd shows the cause.
- **Reload:** scmctl reload sends SIGHUP. The process validates again and applies only keys marked reloadable (thresholds, ease bands, rules). It then publishes I23. An invalid reload keeps the old configuration and logs an error.
- **Find and restore [US-03]:**
  - Every successful load saves a new version. A version becomes *known good* once every component has reported ok (I21) for 10 minutes.
  - scmctl status lists each component's health, so a maintainer can see which service failed.
  - scmctl config rollback restores the latest known-good version (or --to N), checks it, and restarts the affected services. The rollback is logged.

Interface	Input	Output	Errors
config.get (in-process)	key: dotted string	typed value	KeyError at start-up (never at run time, because the schema has defaults)
I23 config.changed (bus)	—	{version: int, changed_keys: string[]}	—
scmctl config check (CLI)	path	exit code 0, or 1 with a list of errors	—
scmctl status (CLI) [US-03]	—	one line per component: name, state (ok / degraded / down), since, detail	exit 2 if busd is unreachable ("health unknown")
scmctl config rollback (CLI) [US-03]	--to N: int, optional (default: latest known good)	exit 0, plus the restored version and the services restarted	exit 1 NO_KNOWN_GOOD · exit 1 INVALID_VERSION (fails the check; nothing changed) · exit 3 RESTART_FAILED (version restored, a service still down)

Glue: Logging and health — owner: Rehman Khalid

Data model: no persistent data in the database. Logs go to journald, capped at 500 MB or 14 days.

Behaviour:

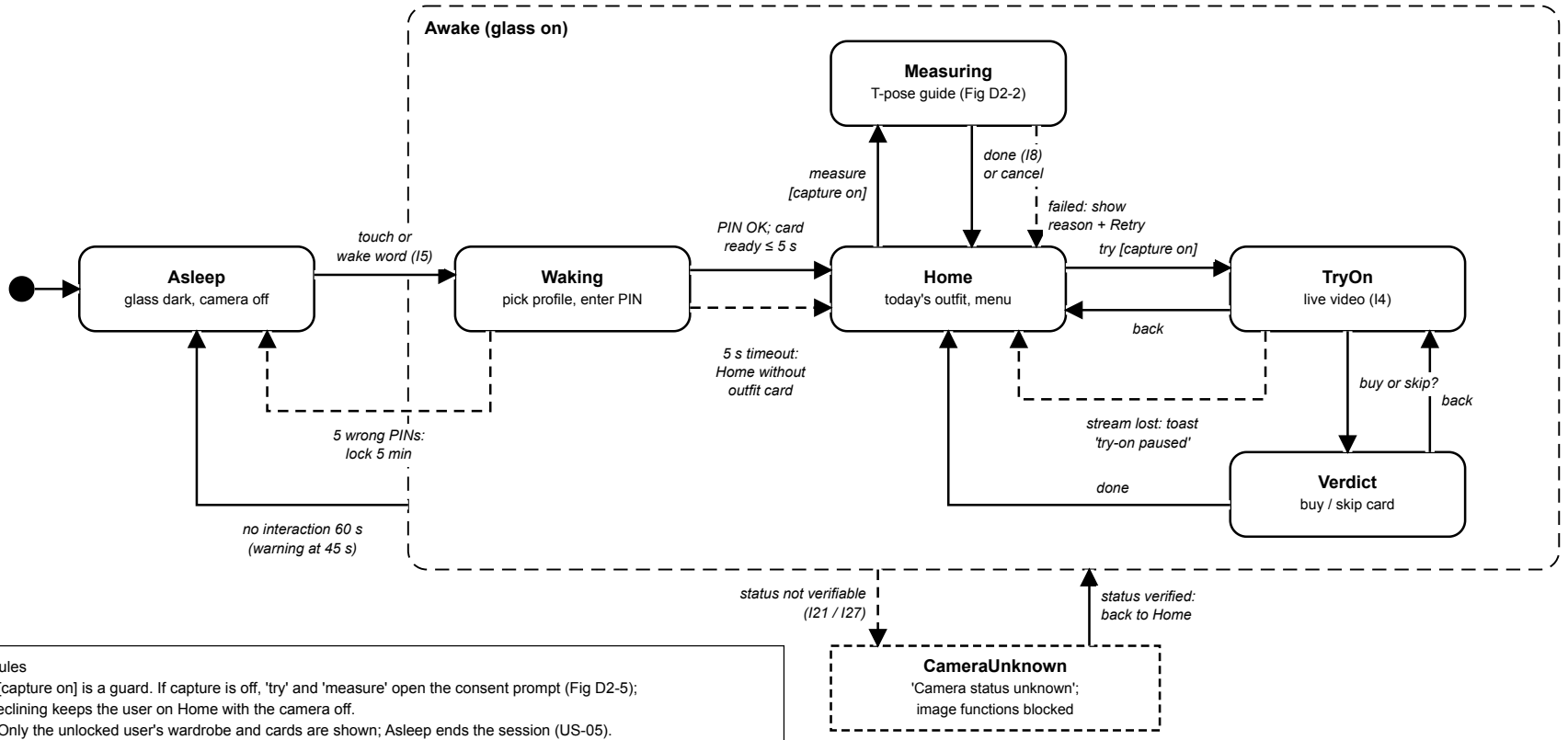
- **Log format:** every component logs JSON lines {ts, level, component, event, msg_id, corr_id, fields}. A corr_id follows one user request across layers.
- **Privacy rule:** never log images, measurements, tokens, or event titles.
- **Health:** every component publishes ops.health every 5 s. busd marks a component DOWN after 15 s of silence.
- **Restarts:** systemd restarts a crashed process, at most 5 times in 10 min.

Interface	Input	Output	Errors
log(event, **fields) (in-process)	event: snake_case string; fields: JSON-safe	—	Fields with blocked names are dropped and counted.
I21 ops.health (bus)	—	{component, state: enum{ok, degraded, down}, detail: ≤ 120 chars, metrics: {fps?, p95_ms?, queue?}}	—
GET /api/v1/health (phone)	—	200 {overall: enum, components: [{name, state, since}]}	401

Smart Closet Mirror — Fig D2-1: Mirror session state machine (L8 Presentation)

Goal: Turn a Gen 1 Mirror into a closet mirror that measures the user with a Kinect, shows their own clothes and shop items on their body, and tells them what to wear today and whether a new piece is worth buying.

Every transition is caused by a Mirror event (I5: touch or wake word), a user action (I17), or a capture-state event (I21, I27). L8 is the only component that changes these states.



Rules
- [capture on] is a guard. If capture is off, 'try' and 'measure' open the consent prompt (Fig D2-5); declining keeps the user on Home with the camera off.
- Only the unlocked user's wardrobe and cards are shown; Asleep ends the session (US-05).
- Socket loss to the WebView is handled inside the Mirror: it reconnects every 2 s (I6).

Legend

State
Error state
Start (first state on power-up)
Transition: trigger / action
Error or timeout transition
Group: an arrow leaving it applies to every state inside

Fig D2-1. Source: Design_Diagrams/D2_fig1_mirror_session.drawio.

Smart Closet Mirror — Fig D2-2: Body measurement state machine (L2 Perception, algorithm A3)

Goal: Turn a Gen 1 Mirror into a closet mirror that measures the user with a Kinect, shows their own clothes and shop items on their body, and tells them what to wear today and whether a new piece is worth buying.

Each state change publishes measure.progress (I25) {state, hint, progress 0–1} so L8 can draw the guide. 'Good frame' = all 20 joints with conf ≥ 0.6.

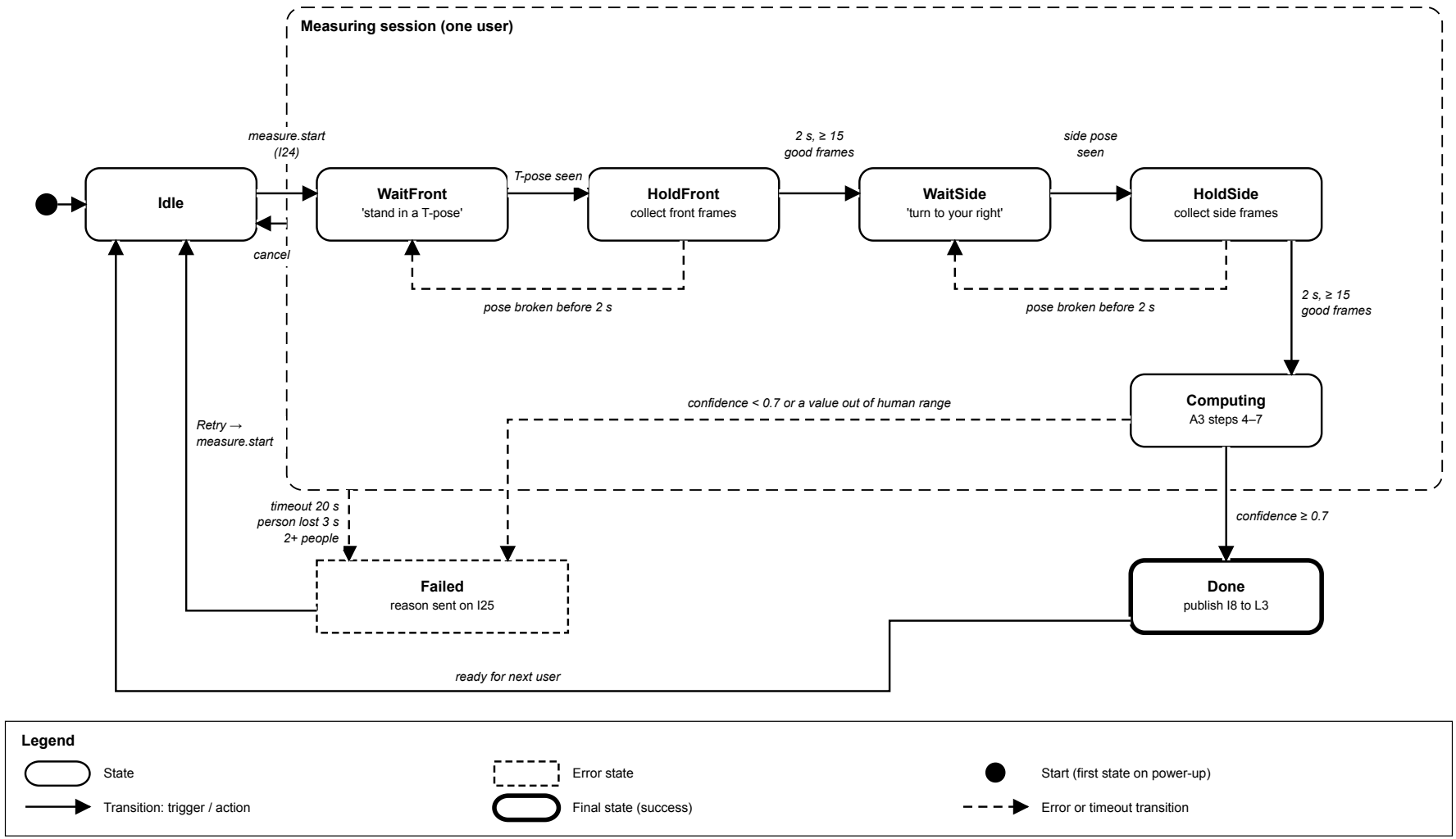


Fig D2-2. Source: Design_Diagrams/D2_fig2_measurement.drawio.

Smart Closet Mirror — Fig D2-3: 'Buy or skip?' decision flow with error paths

Goal: Turn a Gen 1 Mirror into a closet mirror that measures the user with a Kinect, shows their own clothes and shop items on their body, and tells them what to wear today and whether a new piece is worth buying.

Riskiest flow: it crosses L8, L6, L4, L5, L7 and the internet, under a 2 s budget (D0 Flow B). Row 1 runs left to right, row 2 right to left.

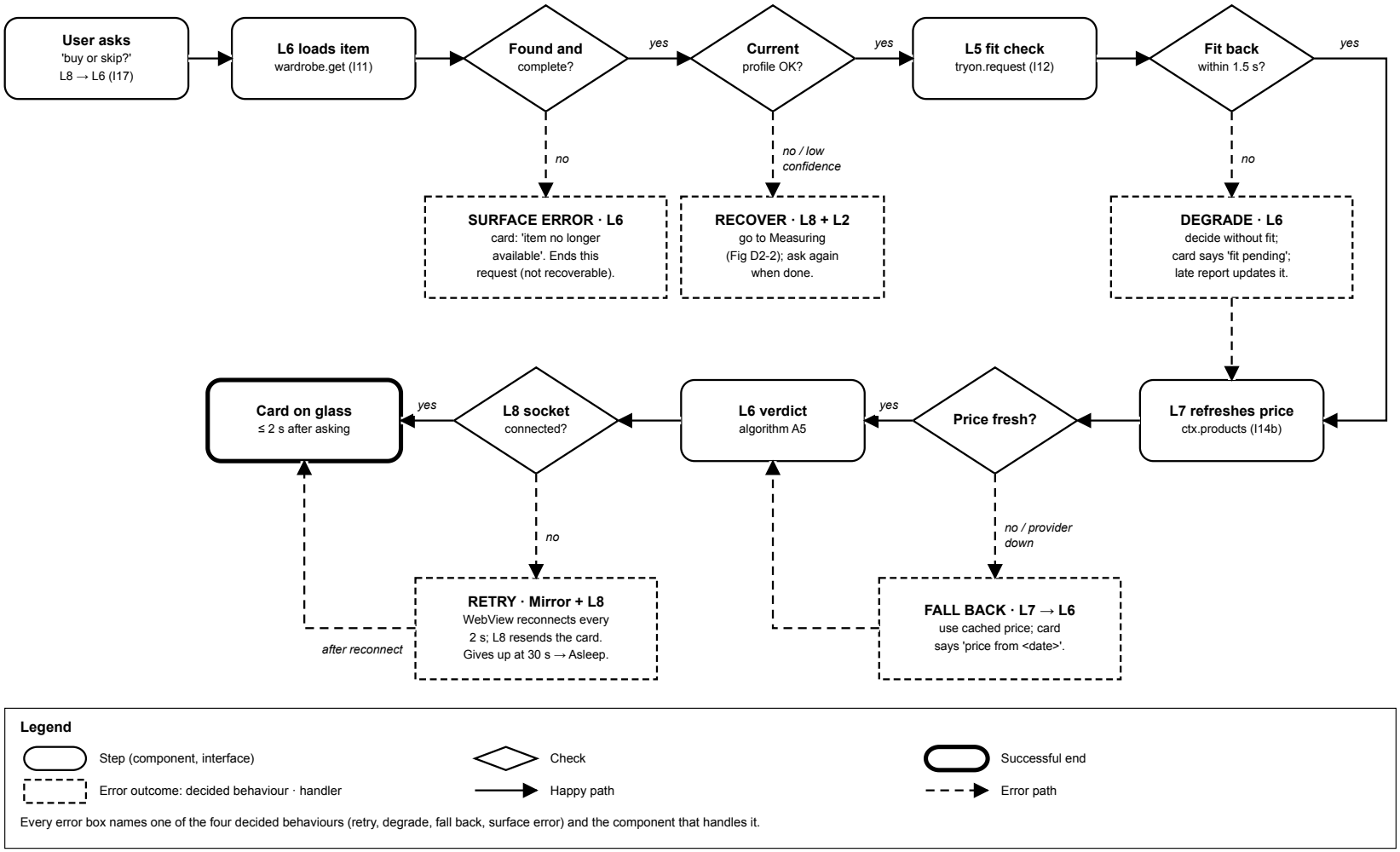


Fig D2-3, the figure required to show error paths. Source: Design_Diagrams/D2_fig3_buy_or_skip_errors.drawio.

Smart Closet Mirror — Fig D2-5: Camera capture consent state machine (L8 + L1, US-04)

Goal: Turn a Gen 1 Mirror into a closet mirror that measures the user with a Kinect, shows their own clothes and shop items on their body, and tells them what to wear today and whether a new piece is worth buying.

L8 owns the prompt and the on-glass indicator; L1 is the only component that can open the Kinect and does so only after capture.enable (I27).

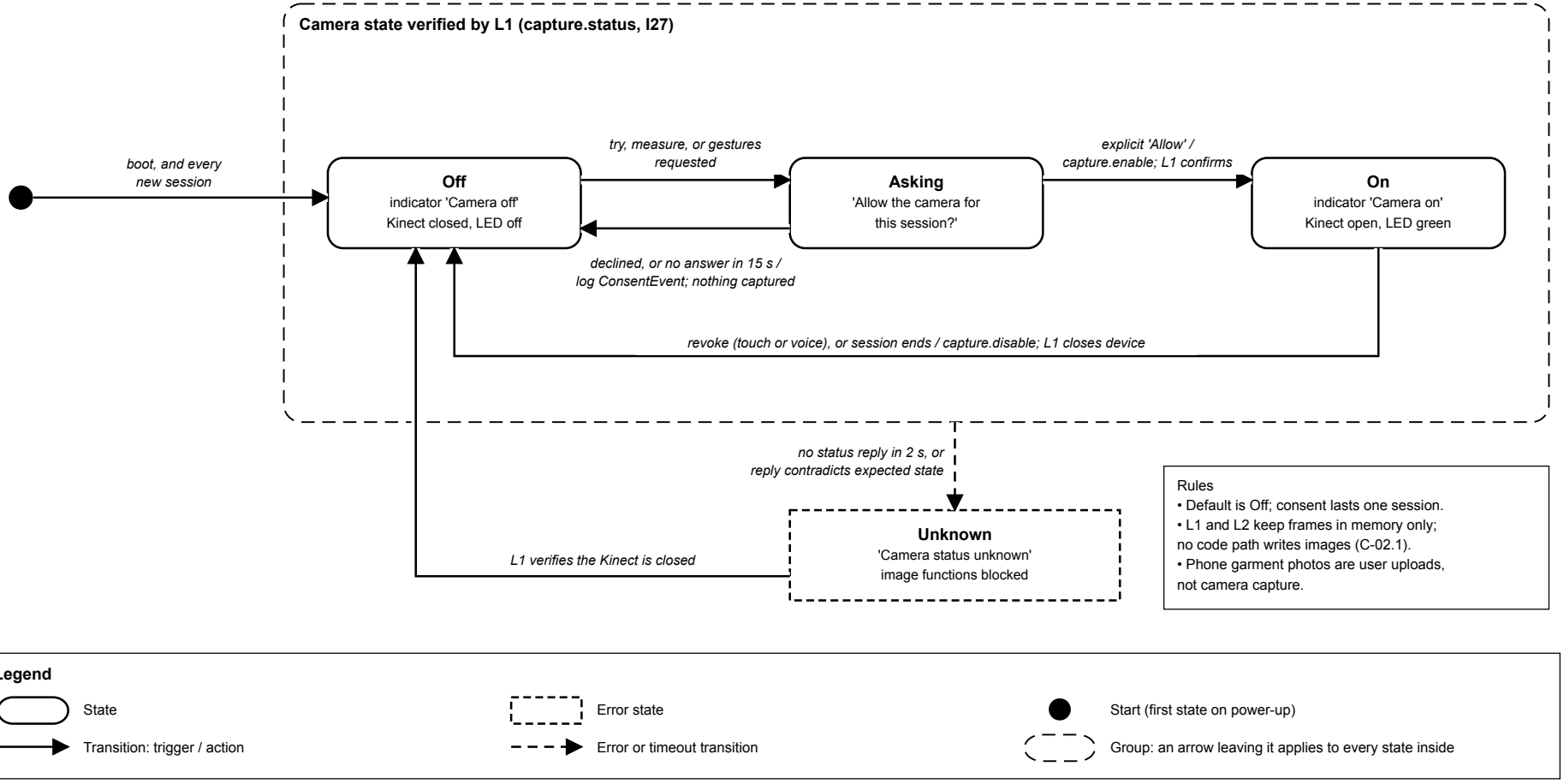


Fig D2-5 [US-04, UC-02]. Source: Design_Diagrams/D2_fig5_camera_consent.drawio.

Smart Closet Mirror — Fig D2-6: 'Recommend an outfit' flow (UC-01, algorithm A4) with error paths

Goal: Turn a Gen 1 Mirror into a closet mirror that measures the user with a Kinect, shows their own clothes and shop items on their body, and tells them what to wear today and whether a new piece is worth buying.

Covers acceptance criteria C-01.1 (main), C-01.2 (weather unavailable), and C-01.3 (incomplete wardrobe). Budget: card ≤ 5 s after the request.

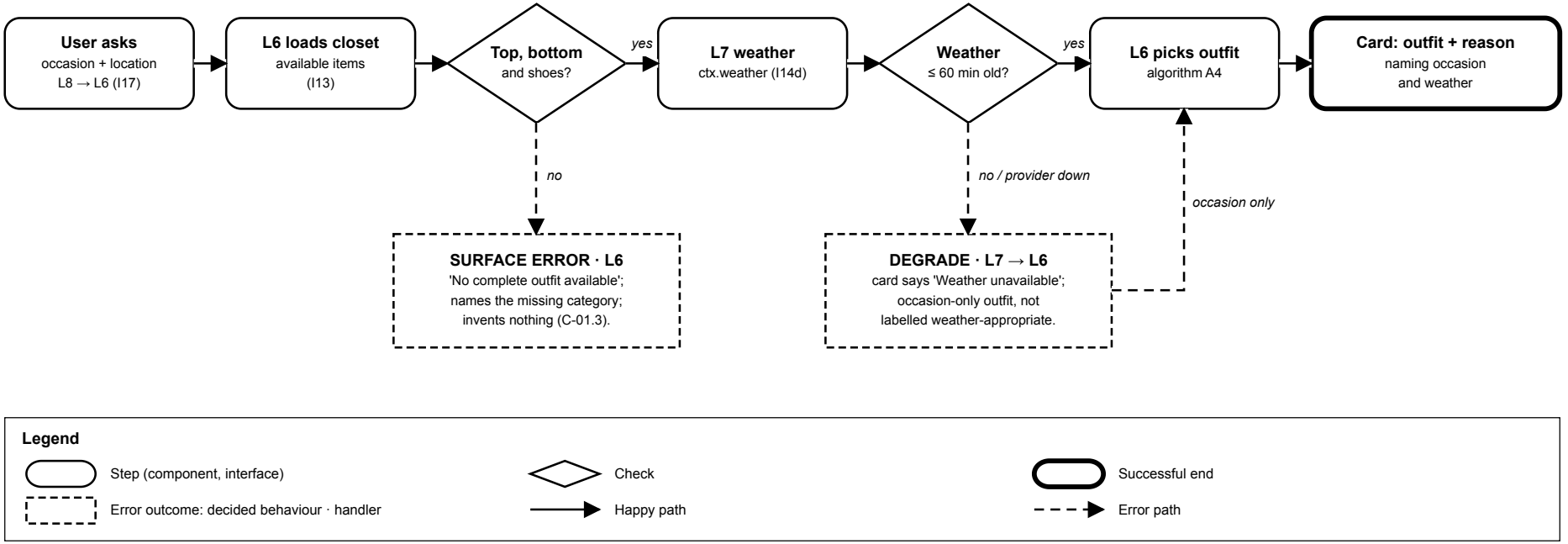


Fig D2-6 [US-01, UC-01]. Source: Design_Diagrams/D2_fig6_outfit_errors.drawio.

Smart Closet Mirror — Fig D2-4: Data model for the components finished in D2

Goal: Turn a Gen 1 Mirror into a closet mirror that measures the user with a Kinect, shows their own clothes and shop items on their body, and tells them what to wear today and whether a new piece is worth buying.

Owners: Device, ProfileLock = Auth · ConsentEvent = L8 · Notification = Notifications · BodyModel = L3 · Decision, StylePreference, SpendEntry = L6 · ProviderAccount, CategoryRule, CacheEntry = L7.

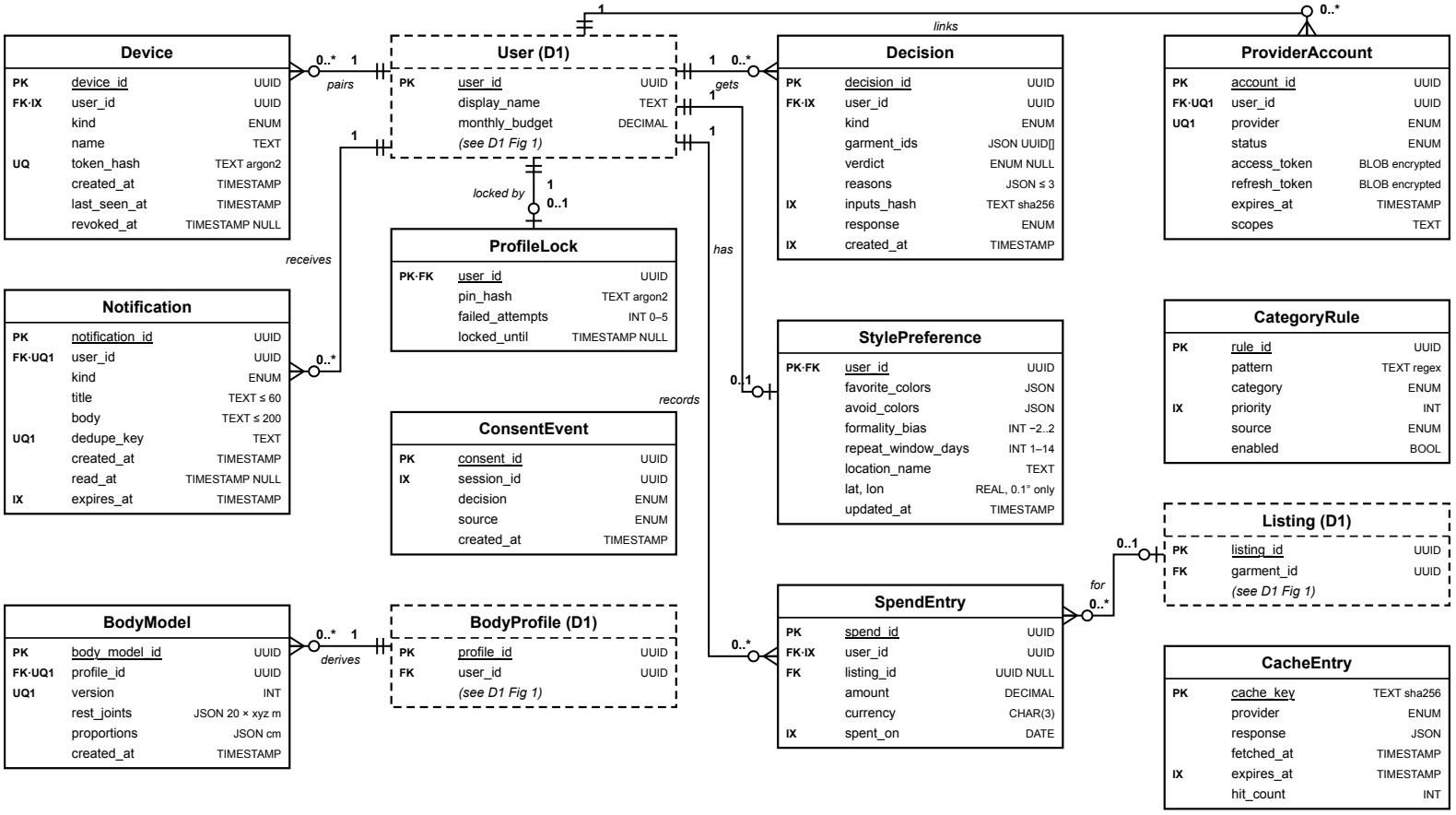


Fig D2-4. Source: Design_Diagrams/D2_fig4_data_model.drawio.

3. Edge cases and error handling

Every component has one row for each of four categories: **B** boundary values, **T** timing and concurrency, **I** bad or missing input, **X** external dependencies. Each row gives the decided behaviour (retry, degrade, fall back, or surface error), the component that handles it, and whether it is recoverable. Figs D2-3 and D2-6 are the error-path diagrams and Fig D2-5 shows the camera-consent failures; their boxes are cited where they apply.

ID	Component	Category	Edge case	Design decision	Handler	Recoverable?
EC-L1-B	L1 Capture	Boundary values	The user stands closer than 0.8 m or farther than 4 m, where the Kinect 360 returns depth 0 for the body.	Pass the frames through unchanged. L2 ignores zero-depth pixels; A3 fails with OUT_OF_RANGE and the hint "Stand 1.5–3 m from the mirror" (Fig D2-2, Failed).	L1, L2 (A3)	Yes: the user steps into range and retries.
EC-L1-T	L1 Capture	Timing / concurrency	RGB and depth callbacks drift more than 20 ms apart, or one stream drops frames, so no pair forms.	Drop the unpaired frame and keep only the newest. No pair for 2 s → Stalled: health DOWN, close the device, reopen every 5 s.	L1	Yes, automatically when pairs resume.
EC-L1-I	L1 Capture	Bad or missing input	capture.enable arrives with an unknown or revoked consent_id, or a second enable arrives while already Streaming.	Surface error: reply NO_CONSENT and stay Disabled, Kinect closed. A repeated enable with a valid consent_id is idempotent and returns the current state.	L1	Yes: L8 asks for consent again (Fig D2-5, Asking).
EC-L1-X	L1 Capture	External dependency	The Kinect is unplugged, loses its 12 V adapter, or reports a USB 2.0 transfer error mid-session.	Report DEVICE_NOT_FOUND or USB_TRANSFER_ERROR on health (DOWN). L8 moves to Unknown, shows "Camera status unknown", and blocks image functions (Fig D2-5, C-02.2).	L1 → L8	Yes, once the device is reconnected (reopen every 5 s).
EC-L2-B	L2 Perception	Boundary values	A gesture lands exactly on a threshold: a swipe of exactly 0.35 m, or confidence exactly 0.8.	Thresholds are fixed as: swipe distance strictly greater than 0.35 m, confidence ≥ 0.8. At most one gesture per second is published (I19), so a borderline double trigger is suppressed.	L2	Yes: the user repeats the gesture.
EC-L2-T	L2 Perception	Timing / concurrency	Pose inference exceeds its 60 ms budget because A2 and NVENC compete for the GPU, so frames queue up.	Degrade: process only the newest FrameBundle and drop older ones (generation check, I7). If p95 stays above 60 ms for 5 s, publish health DEGRADED with p95_ms.	L2	Yes, when load drops.
EC-L2-I	L2 Perception	Bad or missing input	A frame's generation changes during the read (L1 overwrote the slot), or two people are in view during measurement.	Discard the torn frame. During measurement, two or more people → Failed with reason MULTIPLE_PEOPLE and the hint "Only one person in view" (Fig D2-2).	L2	Yes: the next frame, or a retry of measure.start.

EC-L2-X	L2 Perception	External dependency	OWLv2 weights are missing, or the GPU runs out of memory when vision.analyze runs.	Surface error MODEL_UNAVAILABLE on I15; health DEGRADED. Pose runs in a separate model and keeps working. L6 shows "Photo analysis unavailable", and uploads keep the colour the user entered.	L2 → L6	Yes, after the weights are restored or GPU memory is freed.
EC-L3-B	L3 Body	Boundary values	A new measurement differs by exactly 1.0 cm, or a value sits at the edge of the human range.	"Less than 1.0 cm" keeps the old profile, so exactly 1.0 cm creates a new one. Human range: height 120–220 cm, circumferences 50–180 cm, inclusive; anything outside is logged and dropped.	L3	Yes: the user measures again.
EC-L3-T	L3 Body	Timing / concurrency	Two I8 messages for one user arrive close together (a retried or doubled measurement), racing on is_current.	Each update runs in one transaction that re-reads the current profile first; a unique index on (user_id) where is_current = 1 blocks two current profiles. The second message is compared against the first one's result.	L3, data store	Yes, with no user action.
EC-L3-I	L3 Body	Bad or missing input	I8 arrives with confidence below 0.7, missing rest_joints, or the old schema body_measurements/1.	Drop and log (defence in depth, step 1). An unknown schema version is rejected by the bus with BAD_SCHEMA.	L3, message bus	Yes: the user measures again.
EC-L3-X	L3 Body	External dependency	The data store is busy for more than 2 s or full when a new BodyProfile is saved.	Retry once after 2 s on StorageBusy. Still failing, or StorageFull: keep the old profile and send measure.progress failed with the hint "Could not save, please try again".	L3	Yes, once storage is free.
EC-L4-B	L4 Wardrobe	Boundary values	A garment photo sits exactly at a limit: 8 MB, or 400 × 400 px.	Limits are inclusive. 8 MB + 1 byte → 413 PHOTO_TOO_LARGE; under 400 px on either side → 400 VALIDATION_FAILED naming photo (D1 §5).	L4	Yes: the phone re-sends a smaller or larger photo.
EC-L4-T	L4 Wardrobe	Timing / concurrency	Two phones PATCH the same garment at the same time.	The first write wins; the second gets 412 VERSION_CONFLICT from its stale If-Match ETag. The phone re-fetches and shows the current values before the user edits again.	L4	Yes.
EC-L4-I	L4 Wardrobe	Bad or missing input	The same shirt is uploaded twice, a required field is missing, or the cutout and anchors are not ready yet.	409 DUPLICATE_PHOTO returns the existing garment_id; 400 VALIDATION_FAILED lists missing fields. A garment without a cutout stays incomplete: wardrobe.get returns INCOMPLETE, and A4 and try-on skip it.	L4	Yes.

EC-L4-X	L4 Wardrobe	External dependency	The disk is full or the database is locked for more than 2 s during an upload.	Surface error 503 STORAGE_UNAVAILABLE with Retry-After. The phone page keeps the photo and retries after that delay.	L4, data store	Yes, once storage recovers.
EC-L5-B	L5 Try-on	Boundary values	A size chart has a zone missing, or the ease is exactly at a band's min or max.	Bands are inclusive, so ease exactly at min or max is good. A missing zone is unknown; if every zone is unknown, overall is unknown and A5 treats the fit as pending (Fig D2-3, DEGRADE · L6).	L5 (A1)	Yes: the user or L7 adds sizes.
EC-L5-T	L5 Try-on	Timing / concurrency	A second try-on request arrives while a render stream is live, or one A2 frame takes longer than 30 ms.	Reply RENDER_BUSY with the fit still returned (streaming = false). A slow frame is not queued: the renderer draws the newest pose next.	L5	Yes.
EC-L5-I	L5 Try-on	Bad or missing input	tryon.request has more than 4 garments, an unknown garment id, or a profile with confidence below 0.7.	TOO_MANY_GARMENTS · GARMENT_NOT_FOUND (ids in fields) · PROFILE_LOW_CONFIDENCE, which sends L8 to Measuring for a new T-pose (Fig D2-3, RECOVER · L8 + L2).	L5 → L8	Yes.
EC-L5-X	L5 Try-on	External dependency	The GPU context is lost, or NVENC is unavailable when L5 starts.	Fall back: restart the renderer and show plain video meanwhile (D1 §3). No NVENC at start → libx264 software encoding at 20 fps (G5); health DEGRADED.	L5	Yes.
EC-L6-B	L6 Decision	Boundary values	The price exactly equals the budget left, no budget is set, or the closet has exactly 3 items in a category.	A5 rule 2 skips only when the price is above the budget, so equal is within budget. No budget set → rule 2 is skipped and the card says "No budget set". Exactly 3 items is not a gap (rule 4 needs fewer than 3).	L6 (A5)	Yes.
EC-L6-T	L6 Decision	Timing / concurrency	The fit report arrives after the 1.5 s timeout, or the user acts on an old card while a newer one is showing.	Degrade: give the verdict without fit; the card says "fit pending" and the late report updates it (Fig D2-3, DEGRADE · L6). An old view_id gets STALE_VIEW and L6 resends the current view.	L6	Yes.
EC-L6-I	L6 Decision	Bad or missing input	The wardrobe has no available bottom (or top, or shoes), or the item was archived between the listing and the verdict.	Surface error: "No complete outfit available" naming the missing category, never an invented item (Fig D2-6, C-01.3). An archived item gives "Item no longer available" (Fig D2-3, SURFACE ERROR · L6).	L6	No, for that request: the user changes the inventory or picks another item.

EC-L6-X	L6 Decision	External dependency	The weather provider is down or the reading is older than 60 min; the shop price cannot be refreshed.	Degrade: "Weather unavailable" with an occasion-only outfit, not labelled weather-appropriate (Fig D2-6, C-01.2). Fall back to the cached price; the card says "price from <date>" (Fig D2-3).	L7 → L6	Yes, when the provider returns.
EC-L7-B	L7 Connectors	Boundary values	A calendar date outside today ± 7 days, coordinates out of range, a query of 0 or more than 100 characters, or a cached weather reading exactly 60 min old.	BAD_QUERY naming the field. A reading exactly 60 min old is fresh (≤ 60 min); one second older is returned with stale = true.	L7	Yes: the caller fixes the query.
EC-L7-T	L7 Connectors	Timing / concurrency	Many requests empty a provider's rate-limit bucket, or several callers miss the cache with the same key at once.	Identical in-flight requests share one outgoing call, keyed by the cache key. If the bucket stays empty for more than 3 s, reply RATE_LIMITED and the caller degrades.	L7	Yes.
EC-L7-I	L7 Connectors	Bad or missing input	A provider returns malformed JSON, items with missing fields, or a calendar event with an empty title.	Normalisation drops bad items and logs them; a wholly bad response is treated as PROVIDER_UNAVAILABLE (cached data if any). An empty title goes to A6 as casual.	L7	Yes.
EC-L7-X	L7 Connectors	External dependency	A provider returns 5xx or times out, or still returns 401 after one token refresh.	Retry after 1, 2, then 4 s, honouring Retry-After; then return cached data with stale = true. 401 after the refresh → NEEDS_RELINK, and notify.create (I20) sends relink_needed to the user.	L7, Notifications	Yes; a 401 needs the user to relink.
EC-L8-B	L8 Presentation	Boundary values	No answer to the consent prompt at exactly 15 s; a voice command at edit distance 2; more than 4 cards to show.	15 s with no answer counts as Decline (Fig D2-5). Edit distance 2 is rejected with "Sorry, say that again". Cards are cut to 4 with "More on your phone".	L8	Yes.
EC-L8-T	L8 Presentation	Timing / concurrency	The Mirror's WebSocket drops mid-session, or a second Mirror connection arrives.	Retry: the Mirror reconnects every 2 s and L8 resends the full ui.state with a new seq (Fig D2-3, RETRY · Mirror + L8); after 30 s → Asleep. A second connection closes the old one with 4409.	L8, Mirror	Yes.
EC-L8-I	L8 Presentation	Bad or missing input	A wrong PIN, a voice command with confidence below 0.6, or a touch outside any control.	WRONG_PIN shows the attempts left; LOCKED shows when the profile unlocks. Low-confidence voice and stray touches are ignored without changing state.	L8, Auth	Yes (after 5 min if locked).

EC-L8-X	L8 Presentation	External dependency	capture.status gets no reply within 2 s, or the reply contradicts the expected state; mirror.control gets no ack in 1 s.	Move to Unknown: "Camera status unknown" and image functions blocked, never claiming the camera is off (Fig D2-5, C-02.2). mirror.control is retried once, then logged.	L8	Yes, when L1 answers again.
EC-DS-B	Data store	Boundary values	Free disk space falls below 1 GB.	StorageFull: the store goes read-only, health DEGRADED, uploads get 503, and notify.create sends a system notice to every user.	Data store, Logging and health	Yes, after space is freed.
EC-DS-T	Data store	Timing / concurrency	A write lock is held for more than 2 s, for example by the 03:00 backup or two writers.	StorageBusy → HTTP 503 or STORAGE_UNAVAILABLE on the bus. The backup uses SQLite's online backup in small steps, so it never holds the lock for long.	Data store	Yes: the caller retries.
EC-DS-I	Data store	Bad or missing input	A schema migration fails at start-up, or a write breaks a CHECK or foreign-key rule.	Each migration runs in one transaction and rolls back on failure; the process exits with the reason so systemd shows it. A constraint violation writes nothing and returns VALIDATION_FAILED to the caller.	Data store	Yes, after the migration is fixed.
EC-DS-X	Data store	External dependency	The external backup drive is missing at 03:00.	Skip that night's backup, log it, report health DEGRADED with "backup drive missing", and try again the next night.	Data store	Yes, when the drive is reconnected.
EC-BUS-B	Message bus	Boundary values	A message is larger than 1 MB.	Reject with TOO_LARGE; the sender must send a reference instead (frames already travel by shared-memory slot, I7).	Message bus	Yes.
EC-BUS-T	Message bus	Timing / concurrency	A reply arrives after the caller's timeout, or a component misses heartbeats.	The caller discards replies whose corr_id is no longer pending. No heartbeat for 15 s → that component is marked DOWN.	Bus client, busd	Yes.
EC-BUS-I	Message bus	Bad or missing input	A message uses an unknown schema name or version.	Reject with BAD_SCHEMA; drop it and log the sender and schema.	busd	Yes, after the sender is updated.
EC-BUS-X	Message bus	External dependency	busd itself crashes.	systemd restarts it. Clients reconnect with back-off (0.5, 1, 2 s, up to 5 s) and register their services again. Meanwhile scmctl status exits 2 with "health unknown".	systemd, bus client	Yes.

EC-AUTH-B	Authentication	Boundary values	The 5th wrong PIN; a pairing code at 120 s or on its 6th attempt; a PIN of 3 or 7 digits.	The 5th wrong PIN locks the profile for 5 min. An expired code → 410 CODE_EXPIRED; too many attempts → 429. A PIN outside 4–6 digits → 400 VALIDATION_FAILED.	Auth	Yes.
EC-AUTH-T	Authentication	Timing / concurrency	Two phones enter the same pairing code at once, or the session expires while a request is in flight.	The code is used up inside one transaction, so the second phone gets 404 CODE_UNKNOWN. An in-flight request finishes; the next one needs a new unlock.	Auth	Yes.
EC-AUTH-I	Authentication	Bad or missing input	A request asks for another user's data (an owner parameter or a user_id in the body), or sends a malformed token.	Surface error 403 FORBIDDEN (bus: FORBIDDEN); user_id always comes from the session or token (isolation rule). A malformed token → 401 UNAUTHORIZED.	Auth, repositories	No: refused by design.
EC-AUTH-X	Authentication	External dependency	secrets.env or the token encryption key is missing or unreadable at start-up.	Auth refuses to start and names the missing key; HTTP calls get 503 and the Mirror shows "Service unavailable".	Auth, Configuration	Yes, after the secrets file is restored.
EC-NOTIF-B	Notifications	Boundary values	A title over 60 characters, a body over 200, or ttl_h of 0 or 721.	Reject with BAD_INPUT naming the field; text is never silently cut.	Notifications	Yes: the caller fixes the input.
EC-NOTIF-T	Notifications	Timing / concurrency	Many failed calls raise relink_needed for the same user within minutes.	dedupe_key updates the existing unread notification, so the user sees one notice. Marking as read twice has the same effect as once.	Notifications	Yes.
EC-NOTIF-I	Notifications	Bad or missing input	notify.create for an unknown user, or mark-read on another user's notification.	UNKNOWN_USER on the bus; 404 NOT_FOUND over HTTP, which does not reveal that the notification exists.	Notifications	Yes.
EC-NOTIF-X	Notifications	External dependency	The phone page is closed or the phone is off the home network.	No push by design (G4). Notifications wait in the store until their TTL; the glass shows a toast if the user is there.	Notifications	Yes, when the page next opens within the TTL.
EC-CFG-B	Configuration	Boundary values	A reloadable threshold is set outside its allowed range, for example gesture confidence 1.2.	The schema rejects the reload; the old configuration stays and an error is logged.	Configuration	Yes.
EC-CFG-T	Configuration	Timing / concurrency	SIGHUP arrives during a request, or two reloads come back to back; a component crashes within 10 min of a new version.	Each request reads one configuration snapshot; reloads run one at a time. A version is not marked known good unless every component stays ok for 10 min.	Configuration	Yes.

EC-CFG-I	Configuration	Bad or missing input	config.yaml is not valid YAML or is missing a key at start-up; rollback --to N names a version that fails the check.	The process exits naming the bad key; the maintainer runs scmctl config rollback (US-03). A failing --to N → INVALID_VERSION and nothing changes.	Configuration, maintainer	Yes.
EC-CFG-X	Configuration	External dependency	There is no known-good version yet (a fresh install), or a service fails to restart after a rollback.	NO_KNOWN_GOOD, exit 1. RESTART_FAILED, exit 3: the version is restored, and scmctl status names the service still down.	scmctl, maintainer	Partly: a person must fix the failing service.
EC-LOG-B	Logging and health	Boundary values	journald reaches its 500 MB or 14-day cap.	journald drops the oldest entries; health is not affected.	journald	Yes.
EC-LOG-T	Logging and health	Timing / concurrency	A component hangs and goes silent for 15 s without crashing; a crashing process passes 5 restarts in 10 min.	busd marks it DOWN and scmctl status shows it. After 5 restarts systemd stops trying and notify.create sends a system notice.	busd, systemd	Partly: after 5 restarts a maintainer must step in.
EC-LOG-I	Logging and health	Bad or missing input	A log call includes a blocked field (an image, a measurement, a token, or an event title).	The field is dropped and counted; the count appears in health metrics so the leak can be fixed.	log()	Yes.
EC-LOG-X	Logging and health	External dependency	journald is unavailable or the log disk is full.	Log writes fall back to stderr, which systemd captures; logging never blocks a request.	log()	Yes.

4. Requirements traceability table

One row per user story (US-01 to US-05, from User_Stories, Week 4) and per measurable non-functional target (NF-01 to NF-04). Design elements come from the catalogue below; edge-case IDs refer to part 3.

Req ID	Requirement (short)	F / NF	Priority	Design elements	How they satisfy it	Edge cases · figures
US-01	Outfit suggestions from owned clothing, the occasion, and current weather (UC-01; C-01.1–C-01.3)	F	Must	A4 outfit recommendation · I14d ctx.weather (Open-Meteo) · I13 closet index · I14a ctx.calendar_day + A6 · Garment.warmth, Garment.water_resistant · StylePreference (location) · I17 recommend	A4 checks completeness first, so it never suggests an item the user does not own (C-01.3). The weather rules apply only to a reading ≤ 60 min old; otherwise the card says "Weather unavailable" (C-01.2). The reason names the occasion (request or calendar) and the weather (C-01.1).	EC-L6-I, EC-L6-X, EC-L7-B, EC-L7-X · Fig D2-6
US-02	Add, correct, and remove clothing items in my personal inventory	F	Must	POST / GET / PATCH / DELETE /api/v1/garments (D1 §5) · Garment, GarmentSize, GarmentAsset (D1 Fig 1) · I15 vision.analyze (colour) · Garment status incomplete	Add: a phone photo creates a Garment with detected colours. Correct: PATCH with an ETag, so concurrent edits never overwrite each other. Remove: DELETE archives the item, which drops it from A4 and A5 at once. Duplicate photos are refused (pHash).	EC-L4-B, EC-L4-T, EC-L4-I, EC-L4-X, EC-DS-T · D1 Fig 1
US-03	Maintainer can identify a failed service and restore the last working configuration	F	Should	scmctl status · scmctl config rollback · config_history/index.json (known_good) · I21 ops.health · GET /api/v1/health · systemd restarts	scmctl status shows each component's state, so the maintainer can see which service failed. rollback restores the latest known-good configuration (or --to N), checks it, and restarts the affected services.	EC-CFG-I, EC-CFG-X, EC-LOG-T, EC-BUS-X
US-04	Know whether the camera is active and decline image capture (UC-02; C-02.1–C-02.3)	F	Must	I27 capture.enable / disable · capture.status · ConsentEvent · L1 consent gate (Disabled by default) · I5 without camera data · Fig D2-5	The Kinect stays closed until the user explicitly allows it for that session. The glass always shows "Camera on" or "Camera off". Revoking closes the device (C-02.3). If the state cannot be checked, image functions are blocked (C-02.2).	EC-L1-I, EC-L1-X, EC-L8-B, EC-L8-X · Fig D2-5
US-05	Access only my own wardrobe and preferences on a shared mirror	F	Must	ProfileLock · I28 auth.unlock · PUT /api/v1/profile/pin · isolation rule (user_id from the session only) · GET /api/v1/garments owner limited to the caller (403) · Device tokens	Each person unlocks their own profile with a PIN; every query is scoped to that user_id, and asking for anyone else's data returns 403. Cards and toasts show only the unlocked user's items.	EC-AUTH-B, EC-AUTH-I, EC-L8-I, EC-NOTIF-I

NF-01	Weather used in a recommendation is no more than 60 minutes old (C-01.2)	NF	Must	I14d observed_at and stale · L7 cache TTL 15 min · A4 step 3 (60-minute check)	The cache refreshes every 15 min, so a normal reading is at most 15 min old. Anything older than 60 min is flagged stale, and A4 drops the weather rules and says "Weather unavailable".	EC-L7-B, EC-L7-X, EC-L6-X · Fig D2-6
NF-02	Zero image or video files written when consent is declined or after it is revoked (C-02.1, C-02.3)	NF	Must	L1 shared-memory ring buffer (I7, frame_bundle/2) · L2 with no persistent data · logging privacy rule · Fig D2-5 rules	L1 and L2 have no code path that writes frames, and the Kinect is never opened without a valid consent_id. Test: count the files under the data directory before and after a declined session; the count must not change.	EC-L1-I, EC-LOG-I · Fig D2-5
NF-03	Outfit card ≤ 5 s after the request; buy-or-skip verdict ≤ 2 s	NF	Should	A4 cost (< 50 ms) · 3.5 s deadline on I14d (PR-04) · I12 1.5 s timeout · L7 3 s call timeout · Figs D2-3, D2-6	Every slow step has a deadline that leads to a degraded answer instead of a wait: weather after 3.5 s, fit after 1.5 s ("fit pending"), price from cache.	EC-L6-T, EC-L7-T, EC-L5-T · Figs D2-3, D2-6
NF-04	Motion to glass ≤ 250 ms during try-on (goal statement: show clothes on the user's body)	NF	Should	D0 Flow A budget · L2 pose ≤ 60 ms · A2 30 ms · NVENC encode 25 ms · I7 newest-frame-only	Every stage works on the newest frame only and drops older ones, so delay cannot build up. Software encoding (libx264) is the fallback if NVENC fails.	EC-L2-T, EC-L5-T, EC-L5-X

Reverse check: component → requirements

Component	Requirements it serves (IDs)	If none: keep (why) or cut
L1 Capture	US-04, NF-02, NF-04	Keep.
L2 Perception	US-02 (colour detection, I15), US-04 (nothing runs without consent), NF-04	Keep.
L3 Body	NF-04; goal statement ("measures the user")	Keep: no user story names measurement, but try-on and fit (A1) need a body profile (G9).
L4 Wardrobe	US-01, US-02, US-05	Keep.
L5 Try-on	NF-04; goal statement ("shows ... on their body")	Keep: core of the goal statement and of D1 (G9).
L6 Decision	US-01, US-05, NF-01, NF-03	Keep. A5 stays (goal statement: "whether a new piece is worth buying"). A7 moves to stretch scope (PR-02, R9).
L7 Connectors	US-01, NF-01	Keep. I14b and I14c serve A5's price refresh and the stretch-scope A7.
L8 Presentation	US-01, US-04, US-05, NF-03	Keep.
Data store, message bus	US-01 to US-05 (every story stores or moves data through them)	Keep.
Authentication	US-05, US-02 (device token on every phone call)	Keep.

Notifications	US-01 (relink_needed keeps calendar occasions working), US-03 (system notices)	Keep: small and already designed; in-app only for privacy (G4).
Configuration	US-03	Keep.
Logging and health	US-03, NF-02 (privacy rule: no images in logs)	Keep.

Design element catalogue (for citing in parts 3 and 4)

Kind	Elements and where they are defined
Entities	**D1 Fig 1:** User, BodyProfile, Garment, GarmentSize, GarmentAsset, Listing, Outfit, OutfitItem, WearEvent, FitResult. **Fig D2-4:** Device, ProfileLock, ConsentEvent, Notification, BodyModel, Decision, StylePreference, SpendEntry, ProviderAccount, CategoryRule, CacheEntry. **Added attributes:** Garment.warmth, Garment.water_resistant.
Algorithms	**A1** fit evaluation (D1 §3) · **A2** live garment overlay (D1 §3) · **A3** body measurement (L2, Fig D2-2) · **A4** outfit recommendation (L6, Fig D2-6) · **A5** buy-or-skip verdict (L6, Fig D2-3) · **A6** calendar categorisation (L7) · **A7** look from a photo (L6)
HTTP endpoints	**D1 §5:** POST / GET / PATCH / DELETE /api/v1/garments. **D2:** GET /api/v1/decisions · POST /api/v1/spend · PUT /api/v1/preferences · /api/v1/providers/* · POST /api/v1/pair · PUT /api/v1/profile/pin · /api/v1/devices · /api/v1/notifications · GET /api/v1/health
Bus methods and events	wardrobe.get (I11) · tryon.request (I12) · vision.analyze (I15) · ctx.calendar_day / products / images / weather (I14a–d) · capture.enable / disable (I27) · auth.unlock (I28) · measure.start / cancel (I24) · measure.progress (I25) · body.current · capture.status · auth.pair_code · notify.create (I20) · ops.health (I21) · config.changed (I23) · intent.gesture (I19)
Interfaces	I1–I17 (D0 §4, revised above) · I18–I28 (D2)
Figures	D0: block diagram, data flow (Flows A, B). D1: Fig 1 data model. D2: Fig D2-1 session, D2-2 measurement, D2-3 buy-or-skip error paths, D2-4 data model, D2-5 camera consent, D2-6 outfit error paths.
Timing budgets	Motion → glass ≤ 250 ms (D0 Flow A; A2 30 ms, encode 25 ms) · verdict ≤ 2 s (D0 Flow B, Fig D2-3) · outfit card ≤ 5 s after the request (Figs D2-1, D2-6) · weather no older than 60 min (A4, C-01.2) · fit reply ≤ 1.5 s (I12)

5. Design rationale

Ten D1 and D2 decisions, each stating what we chose, what we passed over, and why. Decisions changed by the peer review give the action item ID from part 6. The D0 decision log (D1–D7: brain PC, Kinect 360, H.264 video, single L7, message bus, voice in L8, clip-over bracket) is referred to by number, not repeated.

#	Decision (chosen over the alternative, and why)	Trade-off accepted, and why	Changed by review? (item ID)
R1	We chose SQLite in WAL mode over PostgreSQL because one household's data (about 70,000 rows) fits in one file with no server to run, and WAL lets L5 read while L4 writes (D1 §6).	We accepted one writer at a time and 2 s busy time-outs because the one-writer-per-table rule keeps contention low, and SQLAlchemy leaves a path to PostgreSQL.	No
R2	We chose a 2-D skinned warp (A2) over 3-D cloth simulation or neural try-on because phone photos only give us 2-D garments, and the warp fits A2's 30 ms budget, while neural try-on takes hundreds of ms per frame and has non-commercial licences.	We accepted that garments look flat when the user turns sideways because the mirror is used facing forward, and the fit is still shown through ease.	No
R3	We chose MediaPipe Pose on the Kinect's RGB image over OpenNI/NiTE skeleton tracking because NiTE is no longer maintained or distributed, while MediaPipe is maintained, Apache-2.0, and fits the 60 ms pose budget.	We accepted fusing depth ourselves (median of a 5 × 5 window) and building our own body mask because both are a few lines over data we already have.	No
R4	We chose waking the glass by touch or voice over presence detection from a camera because watching the room before anyone agrees is exactly what US-04 forbids.	We accepted that the mirror does not wake when someone walks up because a bystander's privacy is worth one tap or word.	No
R5	We chose consent per session over remembered consent because a shared mirror cannot tell whether the person in front is the one who agreed before, and per-session consent starts at Off for every visitor (C-02.1).	We accepted one extra prompt per session because it appears only before try-on, measuring, or gestures and is answered with one tap or word.	No
R6	We chose a profile PIN over face recognition or phone unlock because face recognition needs the camera before consent (US-04), and phone unlock fails when the phone is not at hand.	We accepted that a nearby person could watch the PIN (G2) because the shuffled keypad limits it and the PIN protects wardrobe privacy, not payments.	Considered in PR-05; not changed
R7	We chose Open-Meteo over a paid weather API because it is free for non-commercial use and needs no API key, so there is no secret to leak, and we send only coordinates rounded to 0.1°.	We accepted having no service guarantee because the 15 min cache and the "Weather unavailable" path cover outages (C-01.2).	No
R8	We chose a file-based configuration history over storing configuration in the database because configuration must still load and roll back when the database is the broken part (US-03).	We accepted keeping configuration history and database backups in two places because configuration is small and a rollback rarely needs a database change.	No
R9	We chose to move A7 (look from a photo) to stretch scope over building all seven algorithms for the demo because no user story asks for it and it depends on two outside providers (G1).	We accepted that this part of the original idea may not be demonstrated because the time goes to US-01 to US-05, which the stakeholders asked for.	Yes, PR-02

R10	We chose a 3.5 s deadline on the weather call, then an occasion-only outfit, over waiting for L7's full retry sequence (1 + 2 + 4 s) because waiting would break the 5 s outfit budget (NF-03).	We accepted an occasional "Weather unavailable" card on a slow network because the next request uses the weather L7 fetched in the background.	Yes, PR-04

6. Peer design review notes

Reviewer: Austin Heiss. The same table is in Design_Diagrams/Peer_Review_Notes.md. Priorities: must-fix, worth considering, out of scope for now. The must-fix items are applied in this document and marked in parts 1, 2, and 5.

ID	Criterion	Issue	Raised by (team)	Action	Owner	Priority	Due
PR-01	Completeness	The four glue components had only proposed owners, and L8 had no lead, so no one was accountable for the screen every layer feeds.	Austin Heiss	Confirm the glue owners and name Clare Hooper as L8 lead (part 2).	Clare Hooper	Must-fix (done)	Week 7
PR-02	Risk	A7 (look from a photo) has no user story and depends on two providers not yet chosen, so it could take time away from the stories.	Austin Heiss	Move A7 to stretch scope and choose the image and product sources (G1, G9, R9).	Clare Hooper, Jace Schubert	Must-fix (done)	Week 7
PR-03	Clarity	The user stories and D0–D2 use two different project titles.	Austin Heiss	Use "Smart Closet Mirror" everywhere (G8); User_Stories takes it in its next revision.	Arya Garg	Must-fix (done in D2)	Week 8
PR-04	Feasibility	L7's retries on the weather call (1 + 2 + 4 s) can push the outfit card past its 5 s budget.	Austin Heiss	Add a 3.5 s deadline on I14d in A4, then fall back to an occasion-only outfit (NF-03, R10).	Clare Hooper	Must-fix (done)	Week 7
PR-05	Risk	A PIN typed on a shared glass can be watched by someone nearby.	Austin Heiss	Keep the shuffled keypad; evaluate unlocking from the paired phone once the core works (G2, R6).	Jace Schubert	Worth considering	Week 10
PR-06	Risk	A3's ±2 cm accuracy is untested, and the fit check and buy-or-skip verdict rest on it.	Austin Heiss	Tape-measure test on all four team members in the Week 8 hardware spike; tune the calibration factors (G3).	Arya Garg	Worth considering	Week 9
PR-07	Feasibility	Phone notifications only appear while the page is open.	Austin Heiss	Web push sends data off the home network; keep in-app notifications (G4).	Clare Hooper	Out of scope for now	Not scheduled

Summary

Criterion that drew the most feedback: Risk (PR-02, PR-05, PR-06). The review focused on the parts that rest on things we have not yet tested or chosen: measurement accuracy, outside providers, and PIN entry on a shared screen.

Issues we raised while reviewing other teams: none for this round. Austin Heiss reviewed our design one way, and we did not review another team's design.